Spectral Tree Inference from Incomplete Matrices: Fiedler Vector Bounds under Sub-sampling

Itay Gonnen

Abstract

Top-down reconstruction of latent binary phylogenetic trees is fundamentally bottlenecked by the $\Theta(m^2)$ cost of computing pairwise similarities across m leaves. While observing only a sparse, Bernoulli-sampled subset of entries offers a computational workaround, real-world tree imbalance severely degrades the spectral signal, rendering standard balanced-cluster recovery guarantees inadequate. In this work, we frame spectral tree inference as an eigenvector perturbation problem within the matrix-completion regime, showing that the split obtained from the full similarity matrix of an imbalanced latent tree is recovered exactly from the sign pattern of the empirical Fiedler vector computed on a sparse sample. That split is an edge cut of the tree for any topology.

Our primary contribution is a sharp finite-sample guarantee proving that a sampling rate of $p = \Theta(\log m/m)$ suffices for exact recovery of that split at a fixed imbalance. By utilizing a coordinate-wise Neumann-series expansion rather than a standard Davis-Kahan ℓ_2 transfer, we eliminate a spurious $\sqrt{m}$ factor in the entry-wise error bound. Furthermore, we identify the imbalance ratio η as the governing topological parameter and prove it dictates recovery through two distinct failure mechanisms: artificially inflating subspace coherence (rendering recovery statistically unaffordable) and eroding the spectral gap (rendering recovery structurally impossible). This yields an explicit sample complexity dependence of $(1 + \eta)^3$. Simulations on closed-form block models and Kingman coalescent trees with JC69 sequence evolution validate the predicted $\Theta(\log m/m)$ scaling and the isolated degradation mechanics of topological imbalance.

1 Introduction

A latent tree graphical model relates m observed leaves through unobserved internal nodes, and the inference problem is to recover the topology from leaf data alone [2, 24]. Phylogenetics is the canonical instance and supplies our vocabulary: the leaves are extant organisms, the internal nodes their extinct ancestors, the data aligned sequences. Distance-, likelihood- and spectral-based reconstruction are all well developed [23, 13, 17, 20], but datasets now reach tens of thousands of leaves, and the accurate methods are the slow ones.

That pressure produced the top-down spectral family, which is spectral clustering run on a phylogeny. Build a similarity matrix S over the leaves, form the Laplacian $L = D - S$, split the leaf set by the sign pattern of the Fiedler vector of L, recurse on each side, and merge [2, 28, 15]. The split is provably consistent: the two sides are the two halves of a single edge cut — a *split* of the tree [29] — which is what keeps the merge tractable. Both the error and the cost of the recursion land on the top split, since no parent call corrects it and it is the only call that sees all m leaves. Every entry of S is a sequence-pair estimate over a length-ℓ alignment rather than a lookup, so that one call costs $\Theta(m^2)$ estimator evaluations — the cost divide-and-conquer exists to avoid.

This paper asks what happens to the spectrum when that cost is not paid. Compute only a random p-fraction of the pairs, reweight the survivors by $1/p$, and take the Fiedler vector of the

incomplete matrix, in the manner of low-rank matrix completion [8, 9, 7, 22, 18, 5]. Does its sign pattern still recover the split, and how small can p be? There is reason to expect an answer well below 1: under a molecular clock the population matrix has exactly two blocks and the Fiedler vector is constant on each clade (Section 4), so the signal being sampled is extremely concentrated.

Two things keep this from being routine. What decides recoverability from a random subset is how spread the leading subspace is across coordinates, not how large the matrix is [10]; and guarantees for recursive spectral bipartitioning are stated for clusters of comparable size [4, 21], whereas real phylogenies are markedly more imbalanced than standard null models predict [6, 31, 19, 25, 3]. Imbalance pushes both governing quantities the wrong way: it concentrates the split signal onto the smaller clade, raising the sampling cost, and it shrinks the eigenvalue gap that isolates the Fiedler vector, degrading the signal itself. The two failures are distinct — one leaves the eigenvector informative but unaffordable to sample, the other leaves nothing worth sampling — so pooling them shows one curve where there are two mechanisms.

Recovering a split from a sign pattern is a claim about *every* coordinate of an eigenvector, whereas classical perturbation theory controls only its Euclidean norm [11]; passing between the two charges every node as though it were the worst one, which here decides whether a guarantee is satisfiable at all. We therefore work coordinate-wise [12, 14, 1, 27].

Our contributions are:

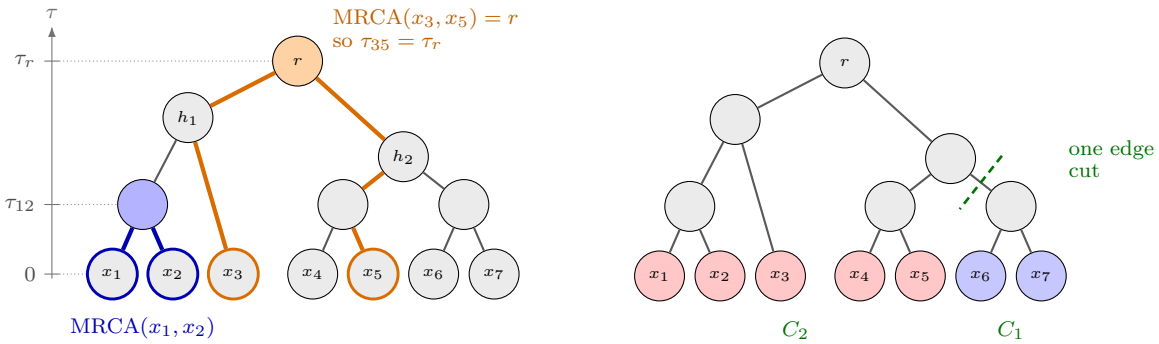
- We make the imbalance ratio η the single parameter through which tree shape reaches the linear algebra, and express both governing spectral quantities in it: the subspace coherence in closed form, and a lower bound on the Fiedler gap (Section 6).
- We prove that a sampling rate of order $\log m/m$ suffices, at fixed imbalance, for the sign pattern of the empirical Fiedler vector to reproduce the full-data split exactly with high probability, with an explicit cubic dependence on η (Section 5).
- The proof works coordinate-wise rather than in Euclidean norm, through a Neumann expansion and a per-node Bernstein bound (Section 7); on this model the Euclidean route costs a further factor of m in the required rate.
- We test the rate in simulation, scoring each sub-sampled split against the one the full matrix returns [26, 16] and stratifying by imbalance rather than pooling (Section 8).

Section 2 fixes the objects: the tree, the substitution model, the similarity graph and its Fiedler vector, and the result that the Fiedler split is always a split of the tree. Section 3 builds the STDR recursion on that result and states the problem — the quadratic cost of its top-level call, the sampling scheme that answers it, and the entry-wise criterion that decides whether the sampled split is the right one. Section 4 collects every hypothesis the analysis carries: the molecular clock, the block model it induces, the imbalance η and the margin ρ. Section 5 states the rate. Section 6 then routes tree shape into the spectrum through η: it identifies which split the population Fiedler vector returns and bounds the two failure modes. Section 7 assembles the proof and Section 8 tests the rate.

Related work is woven into the argument above rather than sectioned off.

2 Background

This section presents the objects the paper works with and the algorithm it accelerates. Section 2.1 sets the tree and the substitution process running on it, Section 2.2 the similarity and distance



(a) Heights and coalescence

(b) Clades and edge cuts

Figure 1: A tree with $m = 7$ terminal nodes. **(a)** Node heights τ increase toward the root, and τ_{ij} is the height of MRCA(x_i, x_j) (see Definition 1). The blue path joins a pair that coalesces early; the orange path joins a pair whose most recent common ancestor is the root, so $\tau_{35} = \tau_r$. **(b)** Removing a single edge splits the terminal nodes in two (Definition 5); the cut shown gives $C_1 = \{x_6, x_7\}$ and $C_2 = \{x_1, \ldots, x_5\}$.

matrices they induce on the terminal nodes, the graph those matrices define, the bipartitions a tree admits, and the result that the Fiedler vector always returns one of them. Section 3 builds the reconstruction algorithm on that result and states the problem this paper solves.

2.1 The generative model

Tree topology

Let $\mathcal{T} = (V, E)$ be a rooted binary phylogenetic tree whose vertex set $V = \mathcal{X} \cup \mathcal{H}$ consists of m observed terminal nodes $\mathcal{X} = \{x_1, \ldots, x_m\}$ and $m-1$ unobserved internal nodes $\mathcal{H} = \{h_1, \ldots, h_{m-1}\}$. We denote by $r \in \mathcal{H}$ the root of the tree. In a binary tree, all terminal nodes have a single neighbor, the internal nodes except the root have three neighbors and the root has two neighbors. In phylogenetics, the terminal nodes represent extant organisms, while the internal nodes represent their common ancestors.

Every edge $e = (h, h') \in E$ is assigned a *branch length* $t_e \geq 0$, which is a measure of the evolutionary divergence between the two adjacent nodes it joins. We represented branch lengths through absolute node heights: each node h is assigned a height $\tau_h \geq 0$ measured upward from the terminal nodes, such that $t_e = t_{(h,h')} = |\tau_h - \tau_{h'}|$.

Definition 1 (Most recent common ancestor)**.** For two termianl nodes x_i, x_j of $\mathcal{X}$, the *most recent common ancestor* MRCA(x_i, x_j) is the node with the smallest value τ among the common ancestors of x_i and x_j [24]. We denote the height of this most recent common ancestor by τ_{ij}.

For a pair of terminal nodes x_i, x_j, let path(x_i, x_j) denote the unique sequence of edges connecting them in $\mathcal{T}$. We define the total length of the path joining them as

$$d_{ij} := \sum_{e \in \text{path}(x_i, x_j)} t_e. \tag{1}$$

Figure 1 illustrates a rooted tree with $m = 7$ terminal nodes.

The substitution process

We assume that the observed sequences are generated according to the common model of sequence evolution, described by a Markov process along the edges of the single latent tree $\mathcal{T}$ (see, e.g., [30, Chapter 1]). Specifically, at each node h in the tree, there is an associated discrete random variable X_h over the alphabet $\{1, \ldots, q\}$. For example, in the common DNA setting, the support of each node consists of $q = 4$ characters representing the basic nucleotides $\{A, C, G, T\}$.

In addition, every edge $e(h, h')$ connecting a pair of adjacent nodes is equipped with transition matrices of size $q \times q$. The sequence generative process begins at the root, where its state is drawn according to some initial distribution over $\{1, \ldots, q\}$. Then, for any parent node $h' \in \mathcal{H}$ connected by an edge to a child node $h \in \mathcal{H}$, the transition matrix $P(h \mid h') \in \mathbb{R}^{q \times q}$ is

$$P(h \mid h')_{ba} = \mathbb{P}(X_h = b \mid X_{h'} = a), \qquad a, b \in \{1, \ldots, q\}. \tag{2}$$

This process propagates from the root to the terminal nodes. The observed sequences of length ℓ for each of the terminal nodes are generated by this process, independently for each location in the sequence.

The GTR model for transition matrices. We assume the GTR model for the transition matrices. Under this model, the sequences mutate along the edges of the tree according to a continuous-time Markov process, with a rate matrix $Q \in \mathbb{R}^{q \times q}$ that satisfies the following four properties:

(i) $Q_{ij} \geq 0$ for $i \neq j$,

(ii) $\sum_j Q_{ij} = 0$ for all i;

(iii) there is a stationary distribution $\pi > 0$ with $\pi Q = 0$; and

(iv) $\operatorname{diag}(\pi)\, Q$ is symmetric.

Conditions (i)–(ii) define Q as the *infinitesimal generator* of a continuous-time Markov chain. This means Q generates the transition probability matrix for any time $t \geq 0$ via the matrix exponential $\exp(Qt)$. Specifically, its off-diagonal entries are the instantaneous rates at which one character is replaced by another, and its rows sum to zero to ensure that $\exp(Qt)$ is a valid stochastic matrix. Condition (iii) says the character frequencies π are left unchanged by the process, and (iv) makes it *reversible* — a path observed forwards in time and the same path observed backwards have the same probability, $\pi_a \left[\exp(Qt)\right]_{ab} = \pi_b \left[\exp(Qt)\right]_{ba}$. Together (i)–(iv) imply that all eigenvalues of Q are non-positive, and hence that $\operatorname{tr}(Q) < 0$.

For two *adjacent* nodes h, h', the substitution matrix over the branch joining them is equal to the matrix exponential of the generator scaled by the branch length,

$$P(h \mid h') = \exp\bigl(Q t_{(h,h')}\bigr), \tag{3}$$

2.2 Similarity, distance and the similarity graph

The substitution model provides a measure of similarity between every pair of nodes in the tree. e first define this pairwise similarity function, and subsequently construct the similarity matrix over the observed terminal nodes.

Definition 2 (Pairwise similarity function)**.** Let $\det(\cdot)$ denote the matrix determinant. The pairwise similarity function between any two nodes h_i and h_j is defined based on the determinants of their transition matrices:

$$S(h_i, h_j) := \sqrt{\det(P(h_i \mid h_j))\, \det(P(h_j \mid h_i))}, \qquad S(h_i, h_i) = 1. \tag{4}$$

Because the determinant is a multiplicative function and the transition matrices P , this similarity measure is also multiplicative along the edges of the tree [2]. Under the GTR model, the transition matrix between two nodes separated by evolutionary distance d_{ij} is $P(x_i \mid x_j) = \exp(Qd_{ij})$. This yields the closed-form expression for any two terminal nodes x_i and x_j:

$$S(x_i, x_j) = \exp\big(\mathrm{tr}(Q)\, d_{ij}\big), \tag{5}$$

with d_{ij} representing the path length defined in Eq. (1). Since the GTR rate matrix satisfies $\mathrm{tr}(Q) < 0$, the similarity decays exponentially with respect to the evolutionary distance: nodes whose most recent common ancestor is recent are highly similar, and this similarity decays geometrically as that ancestor recedes into the past.

Next, we use this function to construct the matrix representation for the observed data.

Definition 3 (Similarity matrix and similarity graph)**.** The *similarity graph* G is the complete weighted graph whose vertices are the m terminal nodes of $\mathcal{T}$ and whose edge (x_i, x_j) carries weight $S(x_i, x_j)$ [2]. Its weighted adjacency matrix is the *similarity matrix* $S \in \mathbb{R}^{m\times m}$, which consists of the pairwise similarities over the terminal nodes. The entries of the matrix S are explicitly mapped as:

$$S_{ij} := S(x_i, x_j), \tag{6}$$

yielding a symmetric matrix with a unit diagonal ($S_{ii} = 1$ for all i).

Next, we define the Laplacian and Fiedler vector of G.

Definition 4 (Laplacian and Fiedler vector)**.** The *combinatorial Laplacian* of the similarity graph is $L := D - S$, where D is diagonal with $D_{ii} = \sum_j S_{ij}$ [2]. Let $0 = \lambda_1 < \lambda_2 \le \lambda_3 \le \cdots \le \lambda_m$ denote the Laplacian's eigenvalues. The *Fiedler vector* $v^{(2)}$ is a unit eigenvector corresponding to the second-smallest eigenvalue λ_2 [15, 28].

To relate the Fiedler vector $v^{(2)}$ to the tree topology, we name the bipartitions of $\mathcal{X}$ that $\mathcal{T}$ admits.

Definition 5 (Clade; split)**.** A *clade* of $\mathcal{T}$ is a node together with all of its descendants, identified with the set of terminal nodes that descend from it [29]. Removing a single edge of $\mathcal{T}$ cuts $\mathcal{X}$ into two parts, a *split* of $\mathcal{T}$ [2]; the part away from the root is a clade, and at the root both parts are.

The splits of $\mathcal{T}$ are therefore its edges, one apiece. Figure 1b illustrates a single cut.
A candidate bipartition is a split of $\mathcal{T}$ exactly when the block of similarities it cuts collapses to rank one [17, Lem. 3.1], which is the precise sense in which the topology of $\mathcal{T}$ is embedded in the structure of S. With the terminal nodes ordered along the split as in Section 2.1, the within-block pairs occupy the two diagonal blocks of S and the cross-block pairs its off-diagonal block.

The following theorem states that thresholding the Fiedler vector of the similarity graph cuts a single edge of $\mathcal{T}$.

Theorem 1 (Consistency of the Fiedler split, [2, Thm. 4.2])**.** *Let G be the similarity graph of a binary tree $\mathcal{T}$ and let v be the Fiedler vector of G. Partition the terminal nodes by the sign pattern of v,*

$$C_1 = \{i :\ v_i \ge 0\}, \qquad C_2 = \{j :\ v_j < 0\}. \tag{7}$$

Then (C_1, C_2) is a split of $\mathcal{T}$.

Theorem 1 makes no assumption on the tree topology or the substitution parameters, and it is the property that underlies the divide-and-conquer algorithm presented in Section 3. We denote by $\Pi(S)$ the bipartition (7) of the Fiedler vector of $L = D - S$.

Algorithm 1 Spectral Top-Down Reconstruction (STDR) for Latent Tree Models. The algorithm recursively partitions terminal nodes using the Fiedler vector of the similarity graph until the subset size falls below m_0. A base method recovers the small subtrees, which are then deterministically merged by identifying optimal placeholder edges via singular vectors of the cross-similarity matrix [2].

Input: $\mathcal{X}$, a set of terminal nodes with associated sequence data of length ℓ; m_0, the maximum partition size for base reconstruction.
Output: $\mathcal{T}$, a rooted binary latent tree spanning the terminal nodes $\mathcal{X}$.
procedure STDR($\mathcal{X}$)
  if $|\mathcal{X}| \le m_0$ **then**
    return BaseReconstruct($\mathcal{X}$) ▷ any accurate method; superlinear in $|\mathcal{X}|$
  $S \leftarrow \big[\, S(x_i, x_j) \,\big]_{x_i, x_j \in \mathcal{X}}$ ▷ $\Theta(|\mathcal{X}|^2)$ pairwise estimates
  $L \leftarrow D - S$ ▷ $D_{ii} = \sum_j S_{ij}$, Definition 4
  $v \leftarrow$ eigenvector of L at λ_2 ▷ the Fiedler vector
  $C_1 \leftarrow \{x_i : v_i \ge 0\}, \quad C_2 \leftarrow \{x_j : v_j < 0\}$ ▷ a split of $\mathcal{T}$, Theorem 1
  $\mathcal{T}_1 \leftarrow \mathrm{STDR}(C_1), \quad \mathcal{T}_2 \leftarrow \mathrm{STDR}(C_2)$
  return Merge($\mathcal{T}_1, \mathcal{T}_2, S(C_1, C_2)$) ▷ join the two subtrees at recovered placeholder edges

3 Problem definition

Because the Fiedler split of the full similarity matrix is always a split of $\mathcal{T}$ (Theorem 1), tree reconstruction becomes a divide-and-conquer recursion. Spectral Top-Down Reconstruction (STDR) [2] forms the similarity matrix over the current node set, splits it by the sign pattern of the Fiedler vector, solves each side independently by any accurate reconstruction method once the subproblem is small enough, and merges the two subtrees along the recovered edge (Algorithm 1).

The recursion is attractive because the expensive part of tree reconstruction is superlinear in the number of terminal nodes, so cutting the problem in half repeatedly reduces the total work sharply, while Theorem 1 guarantees that each cut is a genuine split and the merge is therefore well-posed [2].

The top-level call of Algorithm 1 is the one that governs both the error and the cost. It is the only call that sees all m terminal nodes, and no parent call corrects it: a wrong split at the root cannot be repaired further down. It is also where the similarity matrix is largest.

Forming the full matrix at the top level therefore costs

$$\binom{m}{2} = \Theta(m^2) \text{ pairwise estimates}, \qquad \Theta(m^2 \ell) \text{ character comparisons}, \tag{8}$$

which for $m \sim 10^5$ terminal nodes is on the order of 10^{10} pairwise estimates before the recursion has taken a single step. The quadratic term is exactly the cost that divide-and-conquer was introduced to avoid, and it is paid in full at the root.

This raises the question the paper answers: can the top-down split be computed at almost linear cost, with a guarantee that it is still the same split the full matrix would have returned?

Our answer is to compute only a random subset of the pairs. Following the matrix-completion paradigm [9, 8], we observe entries through a random mask and reweight: let $\Omega \in \{0,1\}^{m \times m}$ be a symmetric mask whose upper-triangular entries are independent Bernoulli(p), $p \in (0,1]$, and form

the empirical similarity by inverse-probability weighting [7],

$$\hat{S}_{ij} = \begin{cases} \frac{1}{p}\,\Omega_{ij}\,S_{ij}, & i \neq j, \\ 1, & i = j. \end{cases} \tag{9}$$

Only the $\Theta(pm^2)$ pairs in the mask are ever estimated, so p is the fraction of the cost in (8) that is actually paid. Under the assumptions of Section 4, a rate $p = \Theta(\log m/m)$ suffices to reproduce the full-data split (Theorem 2), which brings the top-level call down from $\Theta(m^2)$ to $\Theta(m \log m)$ pairwise estimates.

This estimator is unbiased, $\mathbb{E}[\hat{S}] = S$, so $E_L = \hat{L} - L$ is mean zero, where $\hat{L} = \hat{D} - \hat{S}$. The resulting error is missingness rather than additive noise: most pairs are left at zero and the rest rescaled by $1/p$, so it is heavy, structured, and its per-node variance grows as $p \to 0$ (Proposition 4).

Problem. Given the sparsely observed estimate $\hat{S}$ of (9), recover the Fiedler split $\Pi(S)$ of the full matrix from the sign pattern of the Fiedler vector $\hat{v}^{(2)}$ of $\hat{L}$. We ask for the minimal sampling rate p at which $\Pi(\hat{S}) = \Pi(S)$ with high probability. Since $\mathbb{E}[\hat{S}] = S$, the question reduces to an eigenvector-perturbation problem over $E_L = \hat{L} - L$, which the remainder of the paper analyzes. The full-data split $\Pi(S)$ is already a split of $\mathcal{T}$ (Theorem 1), so a rate at which $\Pi(\hat{S}) = \Pi(S)$ is a rate at which the sub-sampled operator returns a genuine split of the tree.

Recovery is, at its core, an *entry-wise* event. Two vectors induce the same sign partition exactly when they agree in sign at every node, so the top-down partition succeeds precisely when the empirical Fiedler vector agrees in sign with its population counterpart coordinate by coordinate, that is, when

$$\left\| \hat{v}^{(2)} - v^{(2)} \right\|_\infty < \min_i |v_i^{(2)}|. \tag{10}$$

Condition (10) is the formal recovery event that every bound in the paper is measured against. Its right-hand side is not computable without a model for S; Section 4 supplies one, under which $\min_i |v_i^{(2)}|$ has a closed form (Lemma 1).

The scope of the analysis is the population matrix. The empirical estimator of an individual entry concentrates around S_{ij} at rate $\mathcal{O}(1/\sqrt{\ell})$ [2], and the structural conditions of Section 4 are stated as properties of the population S, whose observable proxy is that estimator.

4 Modelling assumptions

In this section we state the two hypotheses the analysis carries, the block model they put the similarity matrix in, and the two scalars in which the sampling rate of Section 5 is expressed.

Assumption A1 (GTR substitution model)**.** The substitution model of Section 2.1 holds: a single GTR generator Q satisfying (i)–(iv) drives every edge.

Assumption A2 (Molecular clock)**.** Every branch of $\mathcal{T}$ has positive length, and all m terminal nodes are contemporaneous: $\tau_{x_i} = 0$ for every i.

Assumption A1 already removes per-lineage rate variation — one generator, and a branch length that is a difference of heights — so Assumption A2 is what makes $\mathcal{T}$ ultrametric. That is what places S in the model on which spectral bipartitioning is analysed: the *constant block model* (CBM) of Balakrishnan et al. [4], in the localized form used for latent trees by [2].

Definition 6 (Constant block model at a bipartition)**.** A symmetric matrix S with unit diagonal satisfies the *constant block model* at a bipartition $\mathcal{X} = C_1 \cup C_2$ if its cross-block entries share one value $\beta_0 \in (0,1)$ and every within-block entry exceeds it:

$$S_{ij} = \beta_0 \quad (i \in C_1,\ j \in C_2), \qquad \beta_0 < S_{\text{in}}^{\min} \le S_{ij} \le S_{\max} < 1 \quad (i \neq j \text{ in one block}), \tag{11}$$

where $S_{\text{in}}^{\min}$ and $S_{\max}$ are the extreme within-block entries. The blocks themselves are arbitrary.

Proposition 1 (The population similarity is a constant block model at the root split)**.** *Under Assumptions A1 and A2, the population similarity matrix of Definition 3 satisfies Definition 6 at the root split $\mathcal{X} = C_1 \cup C_2$, whose blocks are complementary clades of $\mathcal{T}$, with*

$$\beta_0 = \exp\bigl(2\,\mathrm{tr}(Q)\,\tau_r\bigr), \qquad S_{\text{in}}^{\min} = \exp\bigl(2\,\mathrm{tr}(Q)\,\tau^{\max}\bigr), \tag{12}$$

for τ_r the root height and $\tau^{\max} < \tau_r$ the greatest within-clade MRCA height.

Appendix A proves it. With the terminal nodes ordered along the split as in Section 2.1, the matrix and its Laplacian take the block form

$$S = \begin{pmatrix} S_{C_1} & \beta_0 J_{|C_1|\times|C_2|} \\ \beta_0 J_{|C_2|\times|C_1|} & S_{C_2} \end{pmatrix}, \qquad L = \begin{pmatrix} D_{C_1} - S_{C_1} & -\beta_0 J_{|C_1|\times|C_2|} \\ -\beta_0 J_{|C_2|\times|C_1|} & D_{C_2} - S_{C_2} \end{pmatrix}, \tag{13}$$

with J the all-ones block and the block degrees $D_{C_1} = \mathrm{diag}(S_{C_1}\mathbf{1}_{|C_1|}) + |C_2|\beta_0 I_{|C_1|}$, $D_{C_2} = \mathrm{diag}(S_{C_2}\mathbf{1}_{|C_2|}) + |C_1|\beta_0 I_{|C_2|}$. The diagonal blocks are whatever the two subtrees make them; only the off-diagonal block is pinned, and that is the single structural fact the spectral analysis of Section 6 runs on.

Two scalars carry the split through the analysis: how unevenly it divides the terminal nodes, and how sharply within-block similarities exceed cross-block ones.

Definition 7 (Imbalance ratio and structural margin)**.** For block sizes $|C_1| \le |C_2|$ with $|C_1| + |C_2| = m$, the *imbalance ratio* is

$$\eta := \frac{|C_2|}{|C_1|} \ \ge\ 1. \tag{14}$$

With β_0 and $S_{\text{in}}^{\min}$ as in (11), the *structural margin* is the amount by which the weakest within-block similarity clears the cross-block baseline,

$$\rho := S_{\text{in}}^{\min} - \beta_0 \ >\ 0. \tag{15}$$

A balanced split has $\eta = 1$, while caterpillar-like lineages reach $\eta = \mathcal{O}(m)$. The pair plays the role of the two constants of Balakrishnan et al. [4]: η is their balanced-cluster constant and ρ their separation gap. Because Definition 6 pins every cross-block entry to the single value β_0, the range restriction of that work contributes no further term to ρ here.

Assumptions A1 and A2 are hypotheses on how the data are generated, and they already fix every quantity in (15). What they do not fix is whether those quantities stand in the relation the analysis needs. That relation is the following inequality. It is not an assumption about the data but a property checkable on any given tree, and it is what delimits the trees the guarantee covers; it is the only condition beyond Assumptions A1 and A2 that the main result carries.

Condition C1 (Separation margin)**.** The structural margin strictly dominates the imbalance-weighted cross-block baseline,

$$\rho > \eta\,\beta_0, \qquad \text{equivalently} \qquad S_{\text{in}}^{\min} > (1+\eta)\,\beta_0. \tag{16}$$

Figure 2 shows the Fiedler split on three simulated trees spanning the range of η: the smaller the clade the rule cuts off, the larger the values its terminal nodes carry.

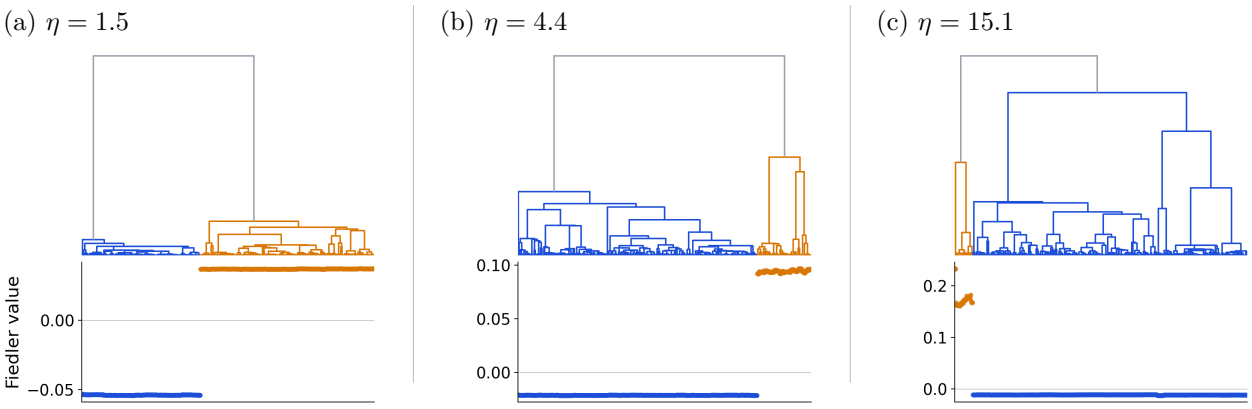


Figure 2: $\Pi(S)$ on three trees, one per panel. Each is drawn from a pool of simulated Kingman trees ($m = 500$ terminal nodes, JC69 sequences of length $\ell = 10^4$) binned by the imbalance its Fiedler split realises. In each panel the tree is on top, its branches coloured by the sign of the Fiedler vector — grey on the one branch that spans both signs — and below it the Fiedler value of each terminal node is plotted at that node's own horizontal position in the tree. The sign pattern cuts a single edge every time, as Theorem 1 requires, and the smaller the clade it cuts off, the larger the values carried by its terminal nodes.

5 Main Result

The entry-wise recovery criterion (10) is what the sampling rate has to buy. This section converts it into a sufficient condition on p for the Laplacian $L = D - S$, stated in the structural quantities of Section 4: the margin ρ of (15) and the imbalance η of (14).

Theorem 2 (Exact recovery of the full-data split)**.** *Let S satisfy the constant block model of Definition 6 with m terminal nodes and imbalance $\eta \geq 1$, let ρ be the structural margin and let $\hat{v}^{(2)}$ be the empirical Fiedler vector — the unit eigenvector belonging to the second-smallest eigenvalue of the sampled Laplacian $\hat{L} = \hat{D} - \hat{S}$, formed from entries observed with probability p as in the observation model* (9)*. Assume Condition C1. For the entry-wise recovery criterion* (10) *to hold with high probability, i.e. $\left\|\hat{v}^{(2)} - v^{(2)}\right\|_\infty < \min_i |v_i^{(2)}|$, it suffices that*

$$p \;>\; \frac{8(1+\eta)^3\,\beta_0^2\,\log m}{m\,(\rho - \eta\,\beta_0)^2} \;=\; \Theta\!\left(\frac{\log m}{m}\right). \tag{17}$$

(17) is a *sufficient* condition. No matching lower bound is proved anywhere in this paper, so the $\Theta(\log m/m)$ is the growth order of that sufficient bound and not a claim that the rate is tight or order-optimal.

6 Topological Scaling and Spectral Gap

This section deals with how the macro-topology of $\mathcal{T}$ fixes its spectrum: the eigenpair that carries the split it carries, the coherence of the leading invariant subspace, and the gap separating λ_2 from λ_3. The whole dependence runs through the single structural parameter η of Definition 7.

Lemma 1 (Spectral decomposition of the imbalanced Laplacian)**.** *Let S satisfy Definition 6 at a bipartition $\mathcal{X} = C_1 \cup C_2$, as Proposition 1 guarantees at the root split under Assumptions A1*

and A2. Then (i) the smallest eigenvalue of L is $\lambda_1 = 0$, simple, with $v^{(1)} = \frac{1}{\sqrt{m}}\mathbf{1}_m$; (ii) $m\beta_0$ is an eigenvalue of L whose eigenvector is piecewise constant on the split, $v_i^{(2)} = +c_1$ for $i \in C_1$ and $v_i^{(2)} = -c_2$ for $i \in C_2$, with $c_1 = \sqrt{|C_2|/(|C_1|m)}$ and $c_2 = \sqrt{|C_1|/(|C_2|m)}$; and (iii) consequently

$$\lambda_2 \leq m\beta_0. \tag{18}$$

If in addition Condition C1 holds, then (iv) (18) holds with equality: $\lambda_2 = m\beta_0$ is simple and $v^{(2)}$ is the Fiedler vector of the similarity graph.

The proof is in Appendix A. Clauses (i)–(iii) are a direct computation, and they use only the constancy of the cross-block, not the floor: the identity $(D_{C_1} - S_{C_1})\mathbf{1}_{|C_1|} = |C_2|\beta_0\mathbf{1}_{|C_1|}$ holds for an arbitrary symmetric diagonal block, because D_{C_1} already carries the row sums of that block. Clause (iv) is where Condition C1 enters, as the statement that $m_{\min}S_{\text{in}}^{\min} > m\beta_0$: it puts the interlacing lower bound on λ_3 used in Lemma 2 strictly above $m\beta_0$, leaving no eigenvalue between 0 and $m\beta_0$. It is what makes $\min_i |v_i^{(2)}| = c_2$ available to the proof of Theorem 2. Since $v^{(2)}$ is non-negative exactly on C_1, the sign rule (7) returns C_1 as its first block, so the labelling of Definition 7 and that of Theorem 1 agree: C_1 is the smaller clade.

6.1 Coherence and the spectral gap under imbalance

Two scalar quantities of the population Laplacian sit behind the constants of (17), and η moves both against recovery. The first is the spectral gap that isolates the Fiedler vector,

$$\Delta\lambda := \lambda_3 - \lambda_2, \tag{19}$$

with λ_2, λ_3 as in Definition 4. The second measures how evenly the leading invariant subspace of L spreads across nodes.

Definition 8 (Subspace coherence)**.** Let $U = [v^{(1)}, v^{(2)}] \in \mathbb{R}^{m\times 2}$ span the invariant subspace of the two smallest eigenvalues of L, with $v^{(1)} = \frac{1}{\sqrt{m}}\mathbf{1}_m$ the trivial flat eigenvector and $v^{(2)}$ the Fiedler vector. The *subspace coherence* is

$$\mu(U) := \tfrac{m}{2} \max_{1\leq i\leq m} \|P_U e_i\|_2^2 = \tfrac{m}{2} \max_{1\leq i\leq m} \big((v_i^{(1)})^2 + (v_i^{(2)})^2\big), \qquad P_U = UU^\top. \tag{20}$$

Coherence is minimal, $\mu(U) = 1$, when the subspace is spread evenly, and grows as it localizes onto few coordinates. Both quantities are fixed by macro-topological features alone: η first inflates $\mu(U)$, then erodes $\Delta\lambda$.

Proposition 2 (Topological scaling of coherence)**.** *Under Definition 6, the tree imbalance η maps deterministically to the subspace coherence,*

$$\mu(U) = \frac{1+\eta}{2}. \tag{21}$$

The identity follows by substituting the Fiedler coordinates of Lemma 1 into (20); the short computation is given in Appendix A. Thus a balanced split attains the optimal coherence $\mu(U) = 1$, and the signal spreads more unevenly across nodes as η grows; the generated regime of Section 8 confirms the identity on data from the full generative pipeline. The identity is reported as a structural diagnostic: since the noise bound of Lemma 3 is obtained from the row norms of S directly (Appendix B), $\mu(U)$ does not itself enter the sampling rate, and the η-dependence of (17) arrives through the margin and the gap instead.

The gap $\Delta\lambda$ of (19) is the quantity the perturbation argument of Section 7 divides by, and the same imbalance that raises $\mu(U)$ drives it toward zero.

Lemma 2 (Spectral-gap lower bound)**.** *Let L be the combinatorial Laplacian of a similarity matrix satisfying the molecular clock, with structural margin ρ as in* (15)*. For $m_{\min} = \min(|C_1|, |C_2|)$ and imbalance η, the spectral gap isolating the Fiedler vector satisfies*

$$\Delta\lambda \geq m_{\min}\rho - \eta\, m_{\min}\, \beta_0. \tag{22}$$

The full algebraic-connectivity derivation is given in Appendix A, building on the eigenpairs of Lemma 1. The bound (22) is purely a CBM statement, controlling the gap through the abstract minimum within-block similarity $S_{\text{in}}^{\min}$ of (11).

6.2 Two failure modes

Proposition 2 and Lemma 2 show that the same parameter η acts on both spectral quantities at once, which produces two distinct ways for top-down recovery to fail; together they bound the admissible region of (m, η). The first is *structural collapse*, $\Delta\lambda \to 0$, which we state here as a corollary of the gap lemma; the second is *statistical infeasibility*, $p > 1$, which is a consequence of Theorem 2. A positive gap requires the structural margin to dominate the imbalance-weighted cross-block boundary, $m_{\min}\rho > \eta\, m_{\min}\beta_0$, i.e. $\eta < S_{\text{in}}^{\min}/\beta_0$ — precisely what Condition C1 records. Translating this ratio into tree distances bounds the imbalance the topology can sustain.

Corollary 1 (Structural imbalance tolerance)**.** *Measure the tree in log-similarity units: by* (5)*, $r(\mathcal{T}) := \max_{i,j}(-\log S_{ij}) = |\mathrm{tr}(Q)| \max_{i,j} d_{ij}$ is its diameter, and $h(\mathcal{T})$ its depth in the same units [2], with $h(\mathcal{T}) < r(\mathcal{T})$. Then a positive spectral gap is possible only if*

$$\eta \leq \mathcal{O}\big(e^{\mathcal{O}(h(\mathcal{T}))-r(\mathcal{T})}\big). \tag{23}$$

In particular, for an average Kingman coalescent tree, where $r(\mathcal{T}), h(\mathcal{T}) = \mathcal{O}(\log m)$ [13, 3], this gives $\eta \leq \mathcal{O}(m^c)$ for a model-dependent constant c.

The proof sketch (substituting the diameter/depth bounds into $\eta < S_{\text{in}}^{\min}/\beta_0$) is given in Appendix A.

So under balanced generative models the gap survives polynomial imbalance. A positive gap is necessary but not sufficient: even when $\Delta\lambda > 0$, the rate (17) grows with η, and the second failure mode is the imbalance past which that demand exceeds $p = 1$.

Corollary 2 (Vacuity of the certificate under imbalance)**.** *Under the hypotheses of Theorem 2, with β_0 and the margin $\rho - \eta\beta_0$ held bounded away from 0 independently of η, the right-hand side of* (17) *exceeds 1 once $\eta = \Omega\big(\sqrt[3]{m/\log m}\big)$; in general the threshold is reached when $\eta^3\beta_0^2/(\rho - \eta\beta_0)^2 = \Omega(m/\log m)$. No admissible sampling rate $p \in (0, 1]$ then satisfies the sufficient condition, so Theorem 2 certifies nothing in that regime. What the theorem stops certifying there is the sampling: Theorem 1 continues to make $\Pi(S)$ a split of $\mathcal{T}$ whatever η does.*

7 Proof Outline

The proof of Theorem 2 compares two quantities evaluated at the same node: the entry-wise error $|\hat{v}_i^{(2)} - v_i^{(2)}|$ and the entry-wise margin $|v_i^{(2)}|$. It runs in three stages — population geometry fixes the margin and noise budget, a resolvent expansion reduces the error at each node to a scalar, and a per-node Bernstein bound controls that scalar — after which imposing (10) yields the rate (17). Each stage is stated as a lemma; the proofs are in Appendices A to C.

Stage 1 (population geometry and noise budget). The same imbalance η sets both sides of (10) at once — it fixes the margin and the spectral gap through the piecewise-constant Fiedler vector of Lemma 1 and the gap bound of Lemma 2, kept positive by Condition C1, while on the other side matrix Bernstein applied to the sampling mask bounds the noise the gap has to absorb.

Lemma 3 (Laplacian spectral error via matrix Bernstein)**.** *Under Definition 6 and the observation model* (9)*, there are constants $\tilde{C}, C_0$ such that if $p \geq C_0 \frac{\log m}{m}$ then with high probability*

$$\|E_L\|_2 \leq \tilde{C}\sqrt{\frac{m \log m}{p}}.$$

The proof is in Appendix B. Compared against the gap bound of Lemma 2, this is what certifies the strong-signal regime $\|E_L\|_2 < \Delta\lambda$ that the next stage assumes. Under imbalance the margin and the gap are extremal at different nodes, which is what makes the final synthesis η-dependent.

Stage 2 (entry-wise reduction). Since (10) is an ℓ_∞ event while Davis–Kahan controls only ℓ_2, and the sole dimension-free transfer between them, $\|\cdot\|_\infty \leq \|\cdot\|_2$, charges every node as though it were the worst one, the reduction has to be coordinate-wise: in the regime of Stage 1 a resolvent (Neumann) expansion of $\hat{v}^{(2)}$ in E_L trades the operator norm for a single scalar per coordinate, at the price of a remainder that is second order in the noise-to-gap ratio.

Lemma 4 (Neumann decomposition of the Fiedler vector)**.** *Let L be symmetric with eigenpairs $\{(\lambda_\ell, v^{(\ell)})\}$, let $\hat{L} = L + E_L$ be a symmetric perturbation, and write $\hat{v}^{(2)} = v^{(2)} + w$. If $\|E_L\|_2 < \Delta\lambda$, then, with Π the spectral projector onto the complement of $v^{(2)}$, w admits the exact resolvent expansion $w = \sum_{k\geq 1}(-1)^k(\Delta\lambda^{-1}\Pi E_L)^k v^{(2)}$, whose first-order term is*

$$w^{(1)} = \sum_{\ell \neq 2} \frac{-\langle v^{(\ell)}, E_L v^{(2)}\rangle}{\lambda_\ell - \hat{\lambda}_2}\, v^{(\ell)}, \tag{24}$$

so that entrywise $w_i = -(E_L v^{(2)})_i/\Delta\lambda + R_i$ with the remainder of second order, $\|R\|_\infty = O\big(\|E_L\|_2^2/\Delta\lambda^2\big)$.

The expansion is due to Eldridge et al. [12], Fan et al. [14] and Abbe et al. [1] rather than to this paper, which is why it is stated as a lemma; the decomposition and its first-order bound are proved in Appendix C.1, and the remainder R separately in Appendix C.3 (Lemma 9), its discard being what confines the argument to the strong-signal regime of Stage 1.

Stage 3 (per-node concentration). The surviving scalar $(E_L v^{(2)})_i$ is a sum of independent, mean-zero increments, each carrying the factor $v_i^{(2)} - v_j^{(2)}$. Every within-block increment therefore vanishes *exactly* — this is a cancellation, not a bound, and it rests on the clock (Assumption A2) making $v^{(2)}$ exactly piecewise constant, so it is the one step that would degrade if the cross-block were only approximately constant. Only cross-block pairs survive, Bernstein's inequality applies node by node, and since this is the only place the sampling rate p enters the final condition, we state the bound in the main text.

Lemma 5 (Per-node concentration)**.** *With scale $R_2 = \beta_0(1+\eta)/(p\sqrt{m\eta})$, for each $i \in C_2$ and $\delta > 0$, with probability $\geq 1-\delta$, $|(E_L v^{(2)})_i| \leq 2\sqrt{2\sigma_2^2 \ln(2/\delta)} + \frac{2}{3}R_2 \ln(2/\delta)$. In the variance-dominated regime ($p = \Theta(\log m/m)$), a union bound over all m nodes at $\delta = \delta_0/m$ gives, with high probability uniformly over i,*

$$|(E_L v^{(2)})_i| \lesssim \sqrt{\frac{8(1+\eta)\beta_0^2 \log(2m/\delta_0)}{\eta\, p}}. \tag{25}$$

The increment representation, the cancellation of the within-block pairs, the per-node variance σ_2^2 and the proof are in Appendix C.2. This is the step that realises the $\sqrt{m}$ saving that the Davis–Kahan route forfeits: the variance of the increment sum is smaller than $\|E_L\|_2^2$ by a factor m, and it is this per-coordinate saving that turns (10) into the rate (17); Appendix D runs the two routes side by side and Remark 2 reports the ratio. The three stages now combine.

Proof of Theorem 2. Step 1 (population geometry). Lemmas 1 and 2 supply the right-hand side of (10) as $\min_i |v_i^{(2)}| = c_2$ and a spectral gap $\Delta\lambda$ that is positive under Condition C1, and Lemma 3 places $\|E_L\|_2$ below it.

Step 2 (entry-wise reduction). Lemma 4 is therefore applicable and gives $w_i = -(E_L v^{(2)})_i/\Delta\lambda + R_i$; discarding R as lower order against c_2 reduces (10) to the per-node condition $|(E_L v^{(2)})_i| < \Delta\lambda |v_i^{(2)}|$ holding at every node i.

Step 3 (concentration). Lemma 5 bounds the left-hand side of that condition uniformly over the m nodes.

Step 4 (synthesis). The per-node condition is tightest on clade C_2, since $\eta \geq 1$. Squaring $\sqrt{8\sigma_2^2 \log m} < \Delta\lambda c_2$ and substituting σ_2^2, $c_2^2 = 1/(m\eta)$ and the gap bound $\Delta\lambda \geq m_{\min}(\rho - \eta\beta_0)$, the factor η cancels and solving for p gives exactly (17) (Proposition 5 and Corollary 4). Under (17) every node satisfies $|w_i| < |v_i^{(2)}|$, hence $\hat{v}_i^{(2)}$ and $v_i^{(2)}$ share a sign; since $v^{(2)}$ is sign-constant on each clade (Lemma 1), the zero-threshold partition of $\hat{v}^{(2)}$ is (C_1, C_2) exactly. □

8 Empirical Results

Two experiments produce the figures below. In the first, S is written down directly from the constant block model of Definition 6 with $S_{\text{in}}^{\min} = 0.9$ and $\beta_0 = 0.05$, so the sampling mask is the only source of error. In the second, dendropy simulates Kingman coalescent trees, evolves JC69 sequences of length $\ell = 10^4$ along their branches, and S is the log-det similarity estimated from the resulting alignment (Assumption A1); tree shape is then off-model and every entry of S carries estimation noise, so this is where the exact cross-block baseline of Assumption A2 and the margin of Condition C1 are put to the test.

Each matrix is sub-sampled independently at every p on a grid and read through the Laplacian L of Theorem 2, rounded by k-means on L_{sym} as that theorem prescribes. A point is an average over 20 independent sub-sampling replicates at that size and imbalance; on the simulated trees the count is set by how many trees the pool holds at that (n, η).

Recovery is scored by the Normalized Mutual Information (NMI) between the split $\Pi(\hat{S})$ returned from the sub-sampled matrix and the reference split $\Pi(S)$ returned by the same operator on the full matrix at $p = 1$. NMI is the standard measure of agreement between two spectral-clustering partitions [16, 21]:

$$\text{NMI}(Y, C) = \frac{2\, I(Y; C)}{H(Y) + H(C)}, \tag{26}$$

with $I(Y; C)$ the mutual information and $H(\cdot)$ the Shannon entropy. NMI attains its maximum value 1 if and only if the two partitions are identical up to relabeling, and decays as they disagree [26, 16]. The quantity being measured is therefore the cost of sub-sampling and nothing else: both partitions come from the same operator on the same tree, and they differ only in how many entries of S were computed.

Corollary 3 (Perfect partition recovery and NMI = 1)**.** *Under the hypotheses of Theorem 2, the partition $\Pi(\hat{S})$ obtained by thresholding $\hat{v}^{(2)}$ at zero satisfies $\Pi(\hat{S}) = \Pi(S)$ with high probability, and therefore* $\text{NMI}(\Pi(\hat{S}), \Pi(S)) = 1$.

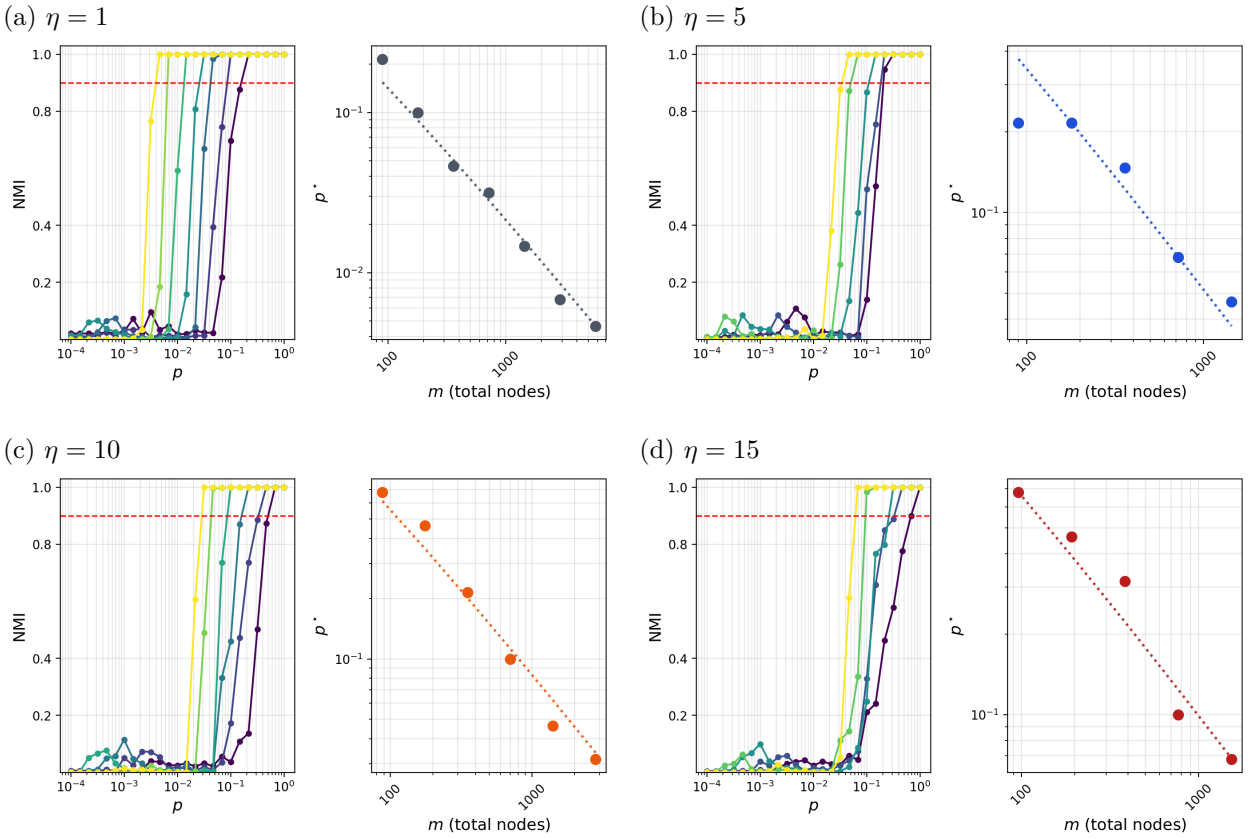


Figure 3: **Similarity route, no estimation noise.** S is the constant block model of Definition 6 in closed form, so the sampling mask is the only error present. The transition sharpens and moves left as m grows, and $\hat{p}^\star$ falls parallel to $\log m/m$ at every imbalance.

Proof. Theorem 2 guarantees $\left\|\hat{v}^{(2)} - v^{(2)}\right\|_\infty < \min_i |v_i^{(2)}|$, so for each node i, $|\hat{v}_i^{(2)} - v_i^{(2)}| < |v_i^{(2)}|$, which forces $\hat{v}_i^{(2)}$ and $v_i^{(2)}$ to share a sign. Thresholding both at zero therefore yields the same partition, $\Pi(\hat{S}) = \Pi(S)$. No property of the tree enters: the argument is node-wise sign agreement between two vectors. Exact agreement gives $I(\Pi(\hat{S}); \Pi(S)) = H(\Pi(\hat{S})) = H(\Pi(S))$, whence by (26) $\mathrm{NMI}(\Pi(\hat{S}), \Pi(S)) = 1$. □

Both figures below share one layout: one pair of panels per imbalance, at $\eta = 1, 5, 10, 15$ throughout. On the left, measured NMI against the sampling rate p, one curve per size, the curves running dark to light (viridis) with increasing size, and the red dashed line at the 0.90 level that defines the empirical threshold $\hat{p}^\star$. On the right, $\hat{p}^\star$ against size on log-log axes, with a dotted reference of the shape (17) predicts, $\log m/m$, fitted to the measured points. Filled markers are measured; open markers are bounds rather than thresholds, a read-off pinned at either end of the p grid, and are excluded from the fit.

Along the similarity route the transition sharpens and shifts left as the size grows, and $\hat{p}^\star$ decays parallel to the reference at every η on both data sets: on the closed-form matrix of Figure 3, and on the simulated phylogenies of Figure 4, where the coherence and gap identities of Proposition 2 and Lemma 2 continue to hold despite off-model tree shapes and JC69 noise. The $\Theta(\log m/m)$ sample complexity of (17) therefore survives finite-sequence estimation noise.

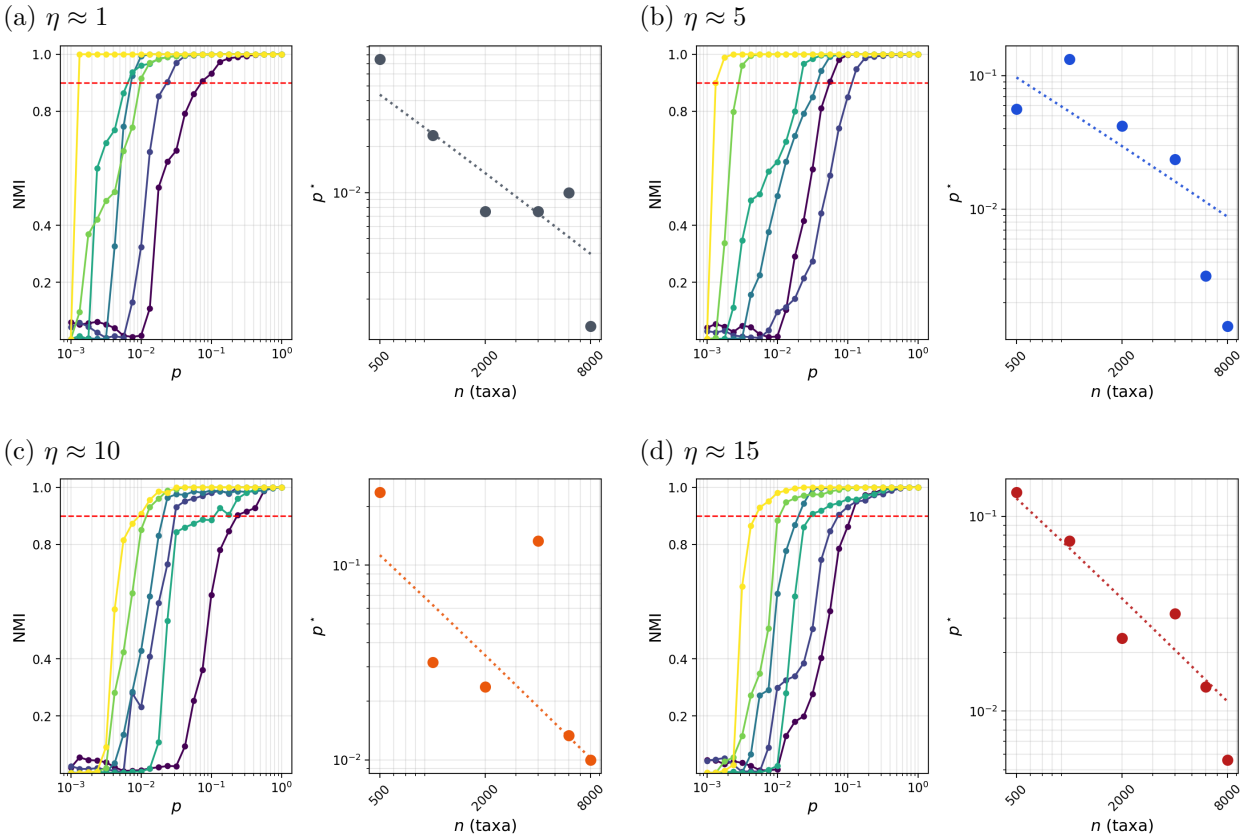


Figure 4: **Similarity route, simulated phylogenies.** Kingman trees carrying JC69 sequences, with S estimated from the alignment rather than written down. The $\log m/m$ decay of Figure 3 survives off-model tree shapes and per-entry estimation noise.

References

[1] Emmanuel Abbe, Jianqing Fan, Kaizheng Wang, and Yiqiao Zhong. Entrywise eigenvector analysis of random matrices with low expected rank. *Annals of Statistics*, 48(3):1452–1474, 2020.

[2] Yariv Aizenbud, Ariel Jaffe, Meng Wang, Amber Hu, Noah Amsel, Boaz Nadler, Joseph T. Chang, and Yuval Kluger. Spectral top-down recovery of latent tree models. *Information and Inference: A Journal of the IMA*, 12(3):2300–2350, 2023.

[3] David J. Aldous. Stochastic models and descriptive statistics for phylogenetic trees, from Yule to today. *Statistical Science*, 16(1):23–34, 2001.

[4] Sivaraman Balakrishnan, Min Xu, Akshay Krishnamurthy, and Aarti Singh. Noise thresholds for spectral clustering. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 24, 2011.

[5] Srinadh Bhojanapalli and Prateek Jain. Universal matrix completion. In *Proceedings of the 31st International Conference on Machine Learning*, volume 32 of *PMLR*, pages 1881–1889, 2014.

[6] Michael G. B. Blum and Olivier François. Which random processes describe the tree of life? A large-scale study of phylogenetic tree imbalance. *Systematic Biology*, 55(4):685–691, 2006.

[7] Emmanuel J. Candès and Yaniv Plan. Matrix completion with noise. *Proceedings of the IEEE*, 98(6):925–936, 2010.

[8] Emmanuel J. Candès and Benjamin Recht. Exact matrix completion via convex optimization. *Foundations of Computational Mathematics*, 9(6):717–772, 2009.

[9] Emmanuel J. Candès and Terence Tao. The power of convex relaxation: Near-optimal matrix completion. *IEEE Transactions on Information Theory*, 56(5):2053–2080, 2010.

[10] Yudong Chen, Srinadh Bhojanapalli, Sujay Sanghavi, and Rachel Ward. Coherent matrix completion. In *Proceedings of the 31st International Conference on Machine Learning (ICML)*, pages 674–682. PMLR, 2014.

[11] Chandler Davis and W. M. Kahan. The rotation of eigenvectors by a perturbation. III. *SIAM Journal on Numerical Analysis*, 7(1):1–46, 1970.

[12] Justin Eldridge, Mikhail Belkin, and Yusu Wang. Unperturbed: spectral analysis beyond Davis-Kahan. In *Proceedings of Algorithmic Learning Theory (ALT)*, pages 321–358. PMLR, 2018.

[13] Péter L. Erdős, Michael A. Steel, László A. Székely, and Tandy J. Warnow. A few logs suffice to build (almost) all trees (I). *Random Structures & Algorithms*, 14(2):153–184, 1999.

[14] Jianqing Fan, Weichen Wang, and Yiqiao Zhong. An ℓ_∞ eigenvector perturbation bound and its application to robust covariance estimation. *Journal of Machine Learning Research*, 18(207):1–42, 2018.

[15] Miroslav Fiedler. A property of eigenvectors of nonnegative symmetric matrices and its application to graph theory. *Czechoslovak Mathematical Journal*, 25(4):619–633, 1975.

[16] Santo Fortunato. Community detection in graphs. *Physics Reports*, 486(3–5):75–174, 2010.

[17] Ariel Jaffe, Noah Amsel, Yariv Aizenbud, Boaz Nadler, Joseph T. Chang, and Yuval Kluger. Spectral neighbor joining for reconstruction of latent tree models. *SIAM Journal on Mathematics of Data Science*, 3(1):113–141, 2021.

[18] Raghunandan H. Keshavan, Andrea Montanari, and Sewoong Oh. Matrix completion from a few entries. *IEEE Transactions on Information Theory*, 56(6):2980–2998, 2010.

[19] J. F. C. Kingman. The coalescent. *Stochastic Processes and their Applications*, 13(3):235–248, 1982.

[20] Elchanan Mossel and Sébastien Roch. Learning nonsingular phylogenies and hidden Markov models. In *Proceedings of the Thirty-Seventh Annual ACM Symposium on Theory of Computing*, pages 366–375, 2005.

[21] Tai Qin and Karl Rohe. Regularized spectral clustering under the degree-corrected stochastic blockmodel. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 26, 2013.

[22] Benjamin Recht. A simpler approach to matrix completion. *Journal of Machine Learning Research*, 12:3413–3430, 2011.

[23] Naruya Saitou and Masatoshi Nei. The neighbor-joining method: a new method for reconstructing phylogenetic trees. *Molecular Biology and Evolution*, 4(4):406–425, 1987.

[24] Charles Semple and Mike Steel. *Phylogenetics*, volume 24 of *Oxford Lecture Series in Mathematics and its Applications*. Oxford University Press, 2003.

[25] Mike Steel and Andy McKenzie. Properties of phylogenetic trees generated by Yule-type speciation models. *Mathematical Biosciences*, 170(1):91–112, 2001.

[26] Alexander Strehl and Joydeep Ghosh. Cluster ensembles—a knowledge reuse framework for combining multiple partitions. *Journal of Machine Learning Research*, 3:583–617, 2002.

[27] Joel A. Tropp. User-friendly tail bounds for sums of random matrices. *Foundations of Computational Mathematics*, 12(4):389–434, 2012.

[28] Ulrike von Luxburg. A tutorial on spectral clustering. *Statistics and Computing*, 17(4):395–416, 2007.

[29] Mark Wilkinson, James O. McInerney, Robert P. Hirt, Peter G. Foster, and T. Martin Embley. Of clades and clans: terms for phylogenetic relationships in unrooted trees. *Trends in Ecology & Evolution*, 22(3):114–115, 2007.

[30] Ziheng Yang. *Molecular Evolution: A Statistical Approach*. Oxford University Press, Oxford, 2014.

[31] George Udny Yule. A mathematical theory of evolution, based on the conclusions of Dr. J. C. Willis, F.R.S. *Philosophical Transactions of the Royal Society of London, Series B*, 213:21–87, 1925. Issue 402–410.

A Proof of the spectral geometry

Every later appeal to a bounded population entry is to (11). No range claim is made for $\hat{S}$: by (9) its observed off-diagonal entries equal S_{ij}/p, which exceed 1 throughout the regime $p = \Theta(\log m/m)$ of interest.

We first verify that the model of Definition 6 is realised at the root split.

Proof of Proposition 1. Under Assumption A2 the tree is ultrametric, $d_{ij} = 2\tau_{ij}$, so (5) reads

$$S_{ij} = \exp\big(2\,\mathrm{tr}(Q)\,\tau_{ij}\big), \tag{27}$$

and S records the coalescence heights of $\mathcal{T}$, one entry per pair.

Let $i \in C_1$ and $j \in C_2$. Every path from C_1 to C_2 traverses the root, so $\mathrm{MRCA}(i,j) = r$ and $\tau_{ij} = \tau_r$ for every cross-block pair; this is purely topological. By (27) all cross-block entries then equal the single value $\beta_0 = \exp(2\,\mathrm{tr}(Q)\tau_r)$, which lies in $(0,1)$ because $\mathrm{tr}(Q) < 0$ and $\tau_r > 0$.

Now let $i \neq j$ lie in the same clade, say C_1, with clade root h_1. Then $\mathrm{MRCA}(i,j)$ is a descendant of h_1, so $\tau_{ij} \le \tau^{\max} \le \max(\tau_{h_1}, \tau_{h_2}) < \tau_r$, the last inequality strict because the two root branches have positive length. As $\mathrm{tr}(Q) < 0$, (27) is strictly decreasing in τ_{ij}, so $S_{ij} \ge \exp(2\,\mathrm{tr}(Q)\tau^{\max}) = S_{\mathrm{in}}^{\min} > \beta_0$. In the other direction, $\mathrm{MRCA}(i,j)$ is a strict ancestor of both i and j for $i \neq j$, so $\tau_{ij} > 0$ and $S_{ij} < 1$; taking the maximum over the finitely many pairs gives $S_{\max} < 1$. Together with $\tau_{ii} = 0$, giving $S_{ii} = \exp(0) = 1$, this is (11) and the stated range. The block degrees in (13) follow by summing the rows of S within each block and adding the constant cross-block contribution, $|C_2|\beta_0$ for a row of C_1 and $|C_1|\beta_0$ for a row of C_2. □

We next prove the spectral decomposition stated in Section 6, which the proof of Lemma 2 then uses.

Proof of Lemma 1(i)–(iii). For any $L = D - S$, $L\mathbf{1} = D\mathbf{1} - S\mathbf{1} = 0$, so 0 is an eigenvalue with $v^{(1)} = \frac{1}{\sqrt{m}}\mathbf{1}_m$; no assumption on S is needed. Every off-diagonal entry of S is at least $\beta_0 > 0$ by (11), so the similarity graph is connected and $\lambda_1 = 0$ is simple.

Step heights. Assume $v^{(2)}$ has the stated form with $c_1, c_2 > 0$. The constraints $\big\|v^{(2)}\big\|_2 = 1$ and $\langle v^{(1)}, v^{(2)}\rangle = 0$ give $|C_1|c_1 = |C_2|c_2$, hence

$$|C_1|\Big(\tfrac{|C_2|}{|C_1|}c_2\Big)^2 + |C_2|c_2^2 = 1 \implies \tfrac{|C_2|m}{|C_1|}c_2^2 = 1 \implies c_2^2 = \tfrac{|C_1|}{|C_2|m}, \quad c_1^2 = \tfrac{|C_2|}{|C_1|m}.$$

Eigenpair verification. Write $v^{(2)} = (c_1\mathbf{1}_{|C_1|},\, -c_2\mathbf{1}_{|C_2|})^\top$ with $|C_1|c_1 = |C_2|c_2$. The only structural input is that the diagonal sub-degrees of (13) already carry the row sums of their own block, $D_{C_1} = \mathrm{diag}(S_{C_1}\mathbf{1}_{|C_1|}) + |C_2|\beta_0 I_{|C_1|}$, so that

$$(D_{C_1} - S_{C_1})\mathbf{1}_{|C_1|} = |C_2|\beta_0\mathbf{1}_{|C_1|}, \qquad (D_{C_2} - S_{C_2})\mathbf{1}_{|C_2|} = |C_1|\beta_0\mathbf{1}_{|C_2|}, \tag{28}$$

identically, for arbitrary symmetric S_{C_1}, S_{C_2}: no constancy of the diagonal blocks is used, and the within-block floor of (11) plays no role here. Using the block form of L with $J_{|C_1|\times|C_2|}\mathbf{1}_{|C_2|} = |C_2|\mathbf{1}_{|C_1|}$, $J_{|C_2|\times|C_1|}\mathbf{1}_{|C_1|} = |C_1|\mathbf{1}_{|C_2|}$ and (28),

$$Lv^{(2)} = \begin{pmatrix} |C_2|\beta_0 c_1\mathbf{1}_{|C_1|} + |C_2|\beta_0 c_2\mathbf{1}_{|C_1|} \\ -|C_1|\beta_0 c_1\mathbf{1}_{|C_2|} - |C_1|\beta_0 c_2\mathbf{1}_{|C_2|} \end{pmatrix},$$

which reduces to $\big(|C_2|\beta_0(c_1{+}c_2)\mathbf{1}_{|C_1|},\, -|C_1|\beta_0(c_1{+}c_2)\mathbf{1}_{|C_2|}\big)^\top$. From $|C_1|c_1 = |C_2|c_2$, $|C_2|(c_1{+}c_2) = mc_1$ and $|C_1|(c_1 + c_2) = mc_2$, so $Lv^{(2)} = m\beta_0\, v^{(2)}$: the pair $(m\beta_0, v^{(2)})$ is an eigenpair, which is claim (ii).

The bound (18). Since $\lambda_1 = 0$ is simple with eigenvector $v^{(1)}$, the Courant–Fischer characterisation of the second eigenvalue reads $\lambda_2 = \min\{x^\top L x : \|x\|_2 = 1,\ x \perp v^{(1)}\}$. The vector $v^{(2)}$ is admissible in that minimisation — it is a unit vector and $\langle v^{(1)}, v^{(2)}\rangle = 0$ by construction — so $\lambda_2 \le \langle v^{(2)}, L v^{(2)}\rangle = m\beta_0 \left\|v^{(2)}\right\|_2^2 = m\beta_0$. □

Nothing above forces λ_2 to *equal* $m\beta_0$; that is an ordering claim, and it is what Condition C1 supplies in clause (iv), proved at the end of this appendix after the interlacing bound it needs.

Proof of Proposition 2. Substituting the Fiedler coordinates of Lemma 1 into (20),

$$\mu(U) = \tfrac{m}{2}\max\Big\{\tfrac{1}{m} + c_1^2,\ \tfrac{1}{m} + c_2^2\Big\} = \tfrac{m}{2}\Big(\tfrac{1}{m} + \tfrac{|C_2|}{m|C_1|}\Big) = \tfrac{1}{2}\Big(1 + \tfrac{|C_2|}{|C_1|}\Big) = \tfrac{1+\eta}{2}.$$ □

Proof of Lemma 2. The argument uses Assumption A2 and Proposition 1, and only the *inequality* (18), $\lambda_2 \le m\beta_0$ — not clause (iv) of Lemma 1, whose proof appeals to the interlacing step below. For λ_3, standard algebraic-connectivity interlacing gives $\lambda_3(L) \ge \min\big(\lambda_2(L_{C_1}), \lambda_2(L_{C_2})\big)$, and applying the Fiedler-eigenvalue formula to each clade Laplacian gives $\lambda_2(L_{C_k}) \ge |C_k|\, S_{\text{in}}^{\min}$ for $k = 1, 2$, with $S_{\text{in}}^{\min}$ the weakest within-block similarity of (11),

$$\lambda_3 \ \ge\ \min\big(\lambda_2(L_{C_1}), \lambda_2(L_{C_2})\big) \ \ge\ \min(|C_1|, |C_2|)\, S_{\text{in}}^{\min} = m_{\min} S_{\text{in}}^{\min}, \tag{29}$$

so $\Delta\lambda = \lambda_3 - \lambda_2 \ge m_{\min} S_{\text{in}}^{\min} - m\beta_0$. Using $S_{\text{in}}^{\min} = \rho + \beta_0$, which is (15), and $m = m_{\min}(1+\eta)$,

$$\Delta\lambda \ge m_{\min}(\rho + \beta_0) - (1+\eta) m_{\min}\beta_0 = m_{\min}\rho - \eta\, m_{\min}\beta_0.$$ □

Proof of Corollary 1. Positivity of the bound in Lemma 2 requires $\eta < S_{\text{in}}^{\min}/\beta_0$. The minimum within-block similarity is bounded by the diameter, $S_{\text{in}}^{\min} \ge e^{-r(\mathcal{T})}$, and the maximum cross-block similarity by the subtree depth, $\beta_0 \propto e^{-\mathcal{O}(h(\mathcal{T}))}$. Substituting into $\eta < S_{\text{in}}^{\min}/\beta_0$ gives (23); the logarithmic scaling $r, h = \mathcal{O}(\log m)$ under Kingman yields $\eta \le \mathcal{O}(e^{\mathcal{O}(\log m)}) = \mathcal{O}(m^c)$. □

With (29) in hand the inequality (18) can be closed.

Proof of Lemma 1(iv). Suppose $\lambda_2 < m\beta_0$. Then $m\beta_0$, an eigenvalue of L by Lemma 1(ii) and equal to neither $\lambda_1 = 0$ nor λ_2, satisfies $m\beta_0 \ge \lambda_3$. The interlacing bound (29) gives $\lambda_3 \ge m_{\min} S_{\text{in}}^{\min}$, so this forces $m_{\min} S_{\text{in}}^{\min} \le m\beta_0$. With $m = m_{\min}(1+\eta)$, Condition C1 reads $S_{\text{in}}^{\min} > (1+\eta)\beta_0$, i.e. $m_{\min} S_{\text{in}}^{\min} > m\beta_0$ — a contradiction. Hence $\lambda_2 = m\beta_0$, and then $\lambda_3 \ge m_{\min} S_{\text{in}}^{\min} > m\beta_0 = \lambda_2$ shows λ_2 is simple, so $v^{(2)}$ is the Fiedler vector up to sign. □

B Proof of the Laplacian sampling error

We prove Lemma 3 by bounding $\|E_L\|_2 \le \|E_D\|_2 + \|E_S\|_2$, where $E_S = \hat{S} - S$ and $E_D = \hat{D} - D$. The only structural fact needed is the entry range (11). By (9) the diagonal is observed exactly, $\hat{S}_{ii} = S_{ii} = 1$, so it carries no noise and only off-diagonal pairs enter the sums below; on those $S_{ij} \le S_{\max} < 1$, giving

$$\max_i \sum_{j \ne i} S_{ij}^2 \ \le\ (m-1)\, S_{\max}^2 \ <\ m\, S_{\max}^2. \tag{30}$$

Proof of Lemma 3. Bounding $\|E_S\|_2$. Write $E_S = \sum_{i<j} Z^{(ij)}$, where $Z^{(ij)}$ is supported on the symmetric off-diagonal pair (i, j) with value $(\frac{1}{p}\Omega_{ij} - 1)S_{ij}$; the summands are independent and

mean zero. The scale is $\left\|Z^{(ij)}\right\|_2 \le \max(\frac{1}{p}-1,1)S_{ij} \le S_{\max}/p =: R$. For the variance proxy, $\mathbb{E}[(Z_{ij}^{(ij)})^2] = \frac{p(1-p)}{p^2}S_{ij}^2 \le \frac{1}{p}S_{ij}^2$, so by (30),

$$\sigma^2 \le \tfrac{1}{p}\max_i \sum_{j\neq i} S_{ij}^2 \le \tfrac{1}{p} m S_{\max}^2.$$

Matrix Bernstein [27], $\mathbb{P}(\|E_S\|_2 \ge t) \le 2m\exp(\frac{-t^2/2}{\sigma^2+Rt/3})$, gives in the variance-dominated regime, with probability $1-\epsilon$, $\|E_S\|_2 \le 2S_{\max}\sqrt{m\ln(2m/\epsilon)/p}$.

Bounding $\|E_D\|_2$. E_D is diagonal with $(E_D)_{ii} = \sum_{j\neq i}(\frac{1}{p}\Omega_{ij}-1)S_{ij}$; a scalar Bernstein with $\sigma_D^2 \le \frac{1}{p}mS_{\max}^2$ and $R_D = S_{\max}/p$ gives $\|E_D\|_2 \lesssim S_{\max}\sqrt{m\ln(2m/\epsilon)/p}$.

Synthesis. The two terms are now of the same order, so

$$\|E_L\|_2 \le \|E_S\|_2 + \|E_D\|_2 \le \tilde{C}\sqrt{\frac{m\log m}{p}},$$

with $\tilde{C}$ absorbing $S_{\max}$ and absolute constants. The bound is uniform in η: the imbalance enters the argument only through the gap $\Delta\lambda$ this noise is compared against, not through the noise itself. □

C Proof of the entry-wise perturbation

C.1 Neumann decomposition (proof of Lemma 4)

Write $\hat{L} = L + E_L$ and $\hat{v}^{(2)} = v^{(2)} + w$, and expand w into a resolvent power series in E_L [12].

Lemma 6 (Exact Neumann decomposition)**.** *Define*

$$w^{(1)} = \sum_{\ell\neq 2} \frac{-\langle v^{(\ell)}, E_L v^{(2)}\rangle}{\lambda_\ell - \hat{\lambda}_2} v^{(\ell)}, \quad w^{(2)} = \sum_{\ell\neq 2} \frac{-\langle v^{(\ell)}, E_L w\rangle}{\lambda_\ell - \hat{\lambda}_2} v^{(\ell)} + \langle v^{(2)}, w\rangle v^{(2)}.$$

Then $w = w^{(1)} + w^{(2)}$ *exactly, and under* $\|E_L\|_2 < \Delta\lambda$ *this is the resolvent series* $w = \sum_{k\ge 1}(-1)^k(\Delta\lambda^{-1}\Pi E_L)^k v^{(2)}$.

Proof. From $\hat{L}\hat{v}^{(2)} = \hat{\lambda}_2\hat{v}^{(2)}$ and $Lv^{(2)} = \lambda_2 v^{(2)}$, $(L-\hat{\lambda}_2 I)w = (\hat{\lambda}_2-\lambda_2)v^{(2)} - E_L v^{(2)} - E_L w$. Projecting onto $v^{(\ell)}$ ($\ell \neq 2$), using $\langle v^{(\ell)}, v^{(2)}\rangle = 0$, gives $\langle v^{(\ell)}, w\rangle = (-\langle v^{(\ell)}, E_L v^{(2)}\rangle - \langle v^{(\ell)}, E_L w\rangle)/(\lambda_\ell - \hat{\lambda}_2)$. Expanding $w = \sum_\ell \langle v^{(\ell)}, w\rangle v^{(\ell)}$ and separating the $\ell = 2$ component recovers the decomposition; iterating on $E_L w$ under $\|E_L\|_2 < \Delta\lambda$ yields the convergent series. □

Lemma 7 (Entry-level first-order bound)**.** *Provided* $\|E_L\|_2 < \Delta\lambda$*, for every node* i*,*

$$|w_i^{(1)}| \le \frac{|(E_L v^{(2)})_i| + \|E_L\|_2\,|v_i^{(2)}|}{\Delta\lambda - \|E_L\|_2}. \tag{31}$$

Proof. By Weyl's inequality $|\lambda_\ell - \hat{\lambda}_2| \ge \Delta\lambda - \|E_L\|_2$ for $\ell \neq 2$. Factoring the denominator, $|w_i^{(1)}| \le |(P_\perp E_L v^{(2)})_i|/(\Delta\lambda - \|E_L\|_2)$ with $P_\perp = I - v^{(2)}(v^{(2)})^\top$. Since $P_\perp x = x - \langle v^{(2)}, x\rangle v^{(2)}$ and $|\langle v^{(2)}, E_L v^{(2)}\rangle| \le \|E_L\|_2 \left\|v^{(2)}\right\|_2^2 = \|E_L\|_2$, $|(P_\perp E_L v^{(2)})_i| \le |(E_L v^{(2)})_i| + \|E_L\|_2\,|v_i^{(2)}|$. □

Remark 1 (Asymmetric correction in the constant block model). The correction $\|E_L\|_2\,|v_i^{(2)}|$ equals $\|E_L\|_2\,c_1$ for $i \in C_1$ and $\|E_L\|_2\,c_2$ for $i \in C_2$. For the binding clade C_2 this is $\|E_L\|_2/\sqrt{m\eta}$, strictly sub-dominant to the recovery target c_2 when $\|E_L\|_2 \ll \Delta\lambda$. Throughout Appendix C.2 we therefore use the leading-order simplification $|w_i^{(1)}| \lesssim |(E_L v^{(2)})_i|/\Delta\lambda$ for $i \in C_2$.

C.2 Per-node Bernstein concentration

Under the leading-order simplification of Remark 1 the entry-wise error at node i is governed by the scalar $(E_L v^{(2)})_i$, which we now represent as a sum of independent increments and concentrate.

Lemma 8 (Laplacian increment representation)**.** *For every node i, $(E_L v^{(2)})_i = \sum_{j\neq i} X_{ij}$ with $X_{ij} = (\frac{\omega_{ij}}{p} - 1)S_{ij}(v_i^{(2)} - v_j^{(2)})$ independent and mean zero.*

Proof. With $(E_S)_{ij} = (\omega_{ij}/p - 1)S_{ij}$, $(E_L)_{ii} = \sum_{k\neq i}(E_S)_{ik}$ and $(E_L)_{ij} = -(E_S)_{ij}$, so $(E_L v^{(2)})_i = \sum_{k\neq i}(E_S)_{ik}v_i^{(2)} - \sum_{j\neq i}(E_S)_{ij}v_j^{(2)} = \sum_{j\neq i}(E_S)_{ij}(v_i^{(2)} - v_j^{(2)})$. □

Proposition 3 (Within-block terms vanish)**.** *Under Definition 6, $v^{(2)}$ is constant on each clade (Lemma 1), so $X_{ij} = 0$ for within-block pairs; only cross-block pairs survive: $(E_L v^{(2)})_i = \sum_{j\in C_1} X_{ij}$ for $i \in C_2$, and $(E_L v^{(2)})_i = \sum_{j\in C_2} X_{ij}$ for $i \in C_1$.*

Proposition 4 (Per-node variance)**.** *Under Definition 6 and the observation model (9), using $(c_1 + c_2)^2 = m/(|C_1||C_2|) = (1+\eta)^2/(m\eta)$,*

$$\sigma_2^2 := \mathrm{Var}\big((E_L v^{(2)})_i\big)\big|_{i\in C_2} = \frac{(1-p)(1+\eta)\beta_0^2}{\eta\, p}, \qquad \sigma_1^2 = \frac{(1-p)(1+\eta)\beta_0^2}{p} = \eta\,\sigma_2^2, \tag{32}$$

so clade C_2 has strictly smaller per-node variance; $\sigma_2^2 = (1-p)\tilde{\rho}(\eta)/p$ with $\tilde{\rho}(\eta) = (1+\eta)\beta_0^2/\eta$.

Proof. For $i \in C_2$ each of the $|C_1|$ surviving terms has $S_{ij} = \beta_0$ and $|v_i^{(2)} - v_j^{(2)}| = c_1 + c_2$, so $\sigma_2^2 = \frac{1-p}{p}(c_1+c_2)^2|C_1|\beta_0^2$; substituting $|C_1| = m/(1+\eta)$ and $(c_1+c_2)^2 = (1+\eta)^2/(m\eta)$ gives the stated value. Clade C_1 follows symmetrically, with its $|C_2|$ cross-block terms. □

Proof of Lemma 5. By Lemma 8 and proposition 3, for $i \in C_2$ the quantity $(E_L v^{(2)})_i = \sum_{j\in C_1} X_{ij}$ is a sum of independent, mean-zero summands with $|X_{ij}| \le \beta_0(c_1+c_2)/p = R_2$ and total variance σ_2^2 (Proposition 4). Bernstein's inequality for bounded independent summands gives the stated two-term deviation bound at confidence $1-\delta$. In the variance-dominated regime $p = \Theta(\log m/m)$ the linear term is of lower order, and a union bound over the m nodes at $\delta = \delta_0/m$ yields (25). □

Proposition 5 (Clade C_2 is binding)**.** *Under Lemma 1, $\min_i |v_i^{(2)}| = |c_2|$, attained on the larger clade C_2. The per-node condition $|(E_L v^{(2)})_i| < \Delta\lambda |v_i^{(2)}|$ requires $p > 8m(1+\eta)\beta_0^2 \log m/\Delta\lambda^2$ on C_2 (the clade-C_1 condition carries an extra $1/\eta$), so since $\eta \ge 1$, clade C_2 binds.*

Proof. Squaring $\sqrt{8\sigma_2^2 \log m} < \Delta\lambda c_2$ and substituting σ_2^2 with $c_2^2 = 1/(m\eta)$: the η cancels, giving $p > 8m(1+\eta)\beta_0^2 \log m/\Delta\lambda^2$. Clade C_1 follows with $c_1^2 = \eta/m$. □

Corollary 4 (Per-node recovery condition)**.** *If $|w_i| < \min_i |v_i^{(2)}|$ for all i then $\mathrm{sign}(\hat{v}^{(2)}) = \mathrm{sign}(v^{(2)})$. Combining Lemma 5 and proposition 5 with the gap bound $\Delta\lambda \ge m(\rho - \eta\beta_0)/(1+\eta)$ (Lemma 2) yields*

$$p > \frac{8(1+\eta)^3\beta_0^2 \log m}{m(\rho - \eta\beta_0)^2} = \Theta\left(\frac{\log m}{m}\right), \tag{33}$$

which is the rate (17) of Theorem 2. This computation is Step 4 of the proof in Section 7.

C.3 The second-order remainder

The bound below is the term discarded at Step 2 of Section 7; it is what restricts the argument to the strong-signal regime $\|E_L\|_2 \ll \Delta\lambda$.

Lemma 9 (Second-order negligibility)**.** $\|w^{(2)}\|_\infty = O\big(\|E_L\|_2^2/\Delta\lambda^2\big)$.

Proof. By Davis–Kahan $\|w\|_2 \le \|E_L\|_2/(\Delta\lambda - \|E_L\|_2)$. Each coefficient $b_\ell = -\langle v^{(\ell)}, E_L w\rangle$ satisfies $|b_\ell| \le \|E_L\|_2 \|w\|_2$, and since $|\lambda_\ell - \hat\lambda_2| \ge \Delta\lambda - \|E_L\|_2$ each summand of $w^{(2)}$ contributes $O(\|E_L\|_2^2/\Delta\lambda^2)$ in ℓ_∞; the $v^{(2)}$ component obeys $|\langle v^{(2)}, w\rangle| = O(\|w\|_2^2) = O(\|E_L\|_2^2/\Delta\lambda^2)$. □

D The Neumann trick — comparison with standard bounds

The recovery criterion (10) constrains $w = \hat v^{(2)} - v^{(2)}$ in the entry-wise sense, $\|w\|_\infty < c_2$, where $c_2 = \min_i |v_i^{(2)}|$ is the entry-wise margin, attained on the binding clade C_2 with $c_2^2 = 1/(m\eta)$, while the sampled-Laplacian noise satisfies $\|E_L\|_2 \le \tilde C\sqrt{m\log m/p}$ (Lemma 3). The two routes below bound the same w, differing only in how they relate the entry-wise constraint to E_L.

The standard (ℓ_2) route. Davis–Kahan controls only the Euclidean norm, $\|w\|_2 \le \|E_L\|_2/(\Delta\lambda - \|E_L\|_2)$. Because the criterion is in the supremum norm, this must be transferred via $\|w\|_\infty \le \|w\|_2$; imposing $\|w\|_2 < c_2$ and substituting the noise bound in the regime $\|E_L\|_2 \ll \Delta\lambda$ yields $p_{\text{naive}} \gtrsim m^2\eta\log m/\Delta\lambda^2$, i.e. $\eta(1+\eta)^2\log m/(\rho-\eta\beta_0)^2$ after the gap bound of Lemma 2.

The entry-wise (Neumann) route. Lemma 4 bounds each coordinate directly, replacing $\|E_L\|_2$ by the scalar $(E_L v^{(2)})_i$ at the tightest coordinate $i \in C_2$. This scalar is a sum of $O(m)$ independent mean-zero contributions concentrating at $\sigma_2^2 = (1-p)(1+\eta)\beta_0^2/(\eta p)$ (Proposition 4), smaller than $\|E_L\|_2^2$ by a factor m; imposing $|w_i| < c_2$ and the gap bound recovers the rate of Corollary 4.

Remark 2 (ℓ_2 vs. entry-wise). Comparing the two rates gives $p_{\text{naive}}/p_{\text{Neumann}} = \Theta\big(m\eta/((1+\eta)\beta_0^2)\big)$, a $\Theta(m)$ reduction for fixed η. The saving originates at the noise level: the operator norm $\|E_L\|_2$ exceeds the individual increment $(E_L v^{(2)})_i$ by $\Theta(\sqrt{m})$, and since p enters both rates as p^{-1} under a square root this squares to $\Theta(m)$. Both rates carry $(\rho-\eta\beta_0)^{-2}$ and so diverge together as $\rho-\eta\beta_0 \to 0$ (the boundary of Condition C1), but the entry-wise rate stays within $p \le 1$ over a region where p_{naive} already exceeds it, so it accommodates more imbalanced trees than the Euclidean route admits.

E Which Split the Fiedler Rule Picks

Two cohorts of 1000 trees, Kingman and birth–death, at $m = 1000$ and $\ell = 10^4$ under Assumptions A1 and A2, with no sub-sampling. The split is taken as Theorem 2 prescribes: k-means on the Fiedler vector of $L(S)$. Per tree we record the imbalance of the returned split, $\eta = |C_2|/|C_1|$ with $|C_1| \le |C_2|$, and whether that split is an edge of the true tree. Figure 5 reports both, per cohort.

Two things follow from Table 1, and both bear on how Section 8 reports its sweeps.

1. **The rule returns an edge of the tree essentially always** — 99.6% on both cohorts — which is Theorem 1 measured under estimation noise rather than on the population matrix.

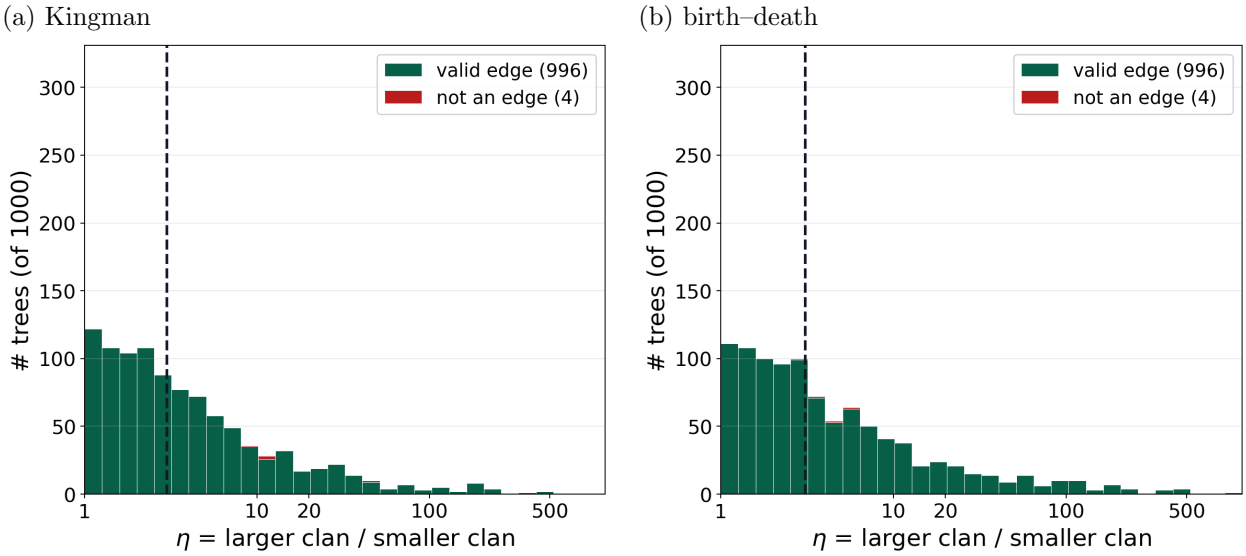


Figure 5: **Imbalance of the returned split, 1000 trees per model.** Green is a genuine single-edge bipartition of the true tree, red is not; dashed line is the median. Both panels share bins and axes.

Table 1: The Fiedler rule on both cohorts, 1000 trees each.

	Kingman	birth–death
valid single-edge split	996	996
median η	3.00	3.08
maximum η	499	999

2. **The imbalance it returns spans three orders of magnitude**, with a median near 3 and a maximum of 499 on Kingman and 999 on birth–death. Since η enters (17) cubically, a sweep that pools trees across this range reports an average over recovery problems of very different difficulty. This is why Section 8 stratifies by η rather than pooling.